

**NAME**

**ovs-appctl** – utility for configuring running Open vSwitch daemons

**SYNOPSIS**

**ovs-appctl** [**--target**=*target* | **-t** *target*] [**--timeout**=*secs* | **-T** *secs*] [**--format**=*format* | **-f** *format*] [**--pretty**] *command* [*arg* ...]

**ovs-appctl --help**

**ovs-appctl --version**

**DESCRIPTION**

Open vSwitch daemons accept certain commands at runtime to control their behavior and query their settings. Every daemon accepts a common set of commands documented under *Common Commands* below. Some daemons support additional commands documented in their own manpages. **ovs-vswitchd** in particular accepts a number of additional commands documented in **ovs-vswitchd(8)**.

The **ovs-appctl** program provides a simple way to invoke these commands. The command to be sent is specified on **ovs-appctl**'s command line as non-option arguments. **ovs-appctl** sends the command and prints the daemon's response on standard output.

In normal use only a single option is accepted:

- **-t** *target* or **--target**=*target*

Tells **ovs-appctl** which daemon to contact.

If *target* begins with / it must name a Unix domain socket on which an Open vSwitch daemon is listening for control channel connections. By default, each daemon listens on a Unix domain socket in the rundir (e.g. /run) named **<program>.<pid>.ctl**, where **<program>** is the program's name and **<pid>** is its process ID. For example, if **ovs-vswitchd** has PID 123, it would listen on **ovs-vswitchd.123.ctl**.

Otherwise, **ovs-appctl** looks in the rundir for a pidfile, that is, a file whose contents are the process ID of a running process as a decimal number, named **target.pid**. (The **--pidfile** option makes an Open vSwitch daemon create a pidfile.) **ovs-appctl** reads the pidfile, then looks in the rundir for a Unix socket named **target.<pid>.ctl**, where **<pid>** is replaced by the process ID read from the pidfile, and uses that file as if it had been specified directly as the target.

- **-T** *secs* or **--timeout**=*secs*

By default, or with a *secs* of **0**, **ovs-appctl** waits forever to connect to the daemon and receive a response. This option limits runtime to approximately *secs* seconds. If the timeout expires, **ovs-appctl** exits with a **SIGALRM** signal.

- **-f** *format* or **--format**=*format*

Tells **ovs-appctl** which output format to use. By default, or with a *format* of **text**, **ovs-appctl** will print plain-text for humans. When *format* is **json**, **ovs-appctl** will return a JSON document. When **json** is requested, but a command has not implemented JSON output, the plain-text output will be wrapped in a provisional JSON document with the following structure:

```
{ "reply-format": "plain", "reply": "$PLAIN_TEXT_HERE" }
```

- **--pretty**

By default, JSON output is printed as compactly as possible. This option causes JSON in output to be printed in a more readable fashion. For example, members of objects and elements of arrays are printed one per line, with indentation. Requires **--format=json**.

## COMMON COMMANDS

Every Open vSwitch daemon supports a common set of commands, which are documented in this section.

### General Commands

These commands display daemon-specific commands and the running version. Note that these commands are different from the **--help** and **--version** options that return information about the **ovs-appctl** utility itself.

- **list-commands**

Lists the commands supported by the target.

- **version**

Displays the version and compilation date of the target.

### Logging Commands

Open vSwitch has several log levels. The highest-severity log level is:

- **off**

No message is ever logged at this level, so setting a logging destination's log level to **off** disables logging to that destination.

The following log levels, in order of descending severity, are available:

- **emer**

A major failure forced a process to abort.

- **err**

A high-level operation or a subsystem failed. Attention is warranted.

- **warn**

A low-level operation failed, but higher-level subsystems may be able to recover.

- **info**

Information that may be useful in retrospect when investigating a problem.

- **dbg**

Information useful only to someone with intricate knowledge of the system, or that would commonly cause too-voluminous log output. Log messages at this level are not logged by default.

Every Open vSwitch daemon supports the following commands for examining and adjusting log levels:

- **vlog/list**

Lists the known logging modules and their current levels.

- **vlog/list-pattern**

Lists logging pattern used for each destination.

- **vlog/set** [*spec*]

Sets logging levels. Without any *spec*, sets the log level for every module and destination to **dbg**. Otherwise, *spec* is a list of words separated by spaces or commas or colons, up to one from each category below:

- A valid module name, as displayed by the **vlog/list** command on **ovs-appctl(8)**, limits the log level change to the specified module.
- **syslog**, **console**, or **file**, to limit the log level change to only to the system log, to the console, or to a file, respectively.
- **off**, **emer**, **err**, **warn**, **info**, or **dbg**, to control the log level. Messages of the given severity or higher will be logged, and messages of lower severity will be filtered out. **off** filters out all messages.

Case is not significant within *spec*.

Regardless of the log levels set for **file**, logging to a file will not take place unless the target application was invoked with the **--log-file** option.

For compatibility with older versions of OVS, **any** is accepted within *spec* but it has no effect.

- **vlog/set PATTERN:destination:pattern**

Sets the log pattern for *destination* to *pattern*. Each time a message is logged to *destination*, *pattern* determines the message's formatting. Most characters in *pattern* are copied literally to the log, but special escapes beginning with **%** are expanded as follows:

- **%A**

The name of the application logging the message, e.g. **ovs-vswitchd**.

- **%B**

The RFC5424 syslog PRI of the message.

- **%c**

The name of the module (as shown by **ovs-appctl --list**) logging the message.

- **%d**

The current date and time in ISO 8601 format (**YYYY-MM-DD HH:MM:SS**).

- **%d{format}**

The current date and time in the specified *format*, which takes the same format as the **template** argument to **strftime(3)**. As an extension, any **#** characters in *format* will be replaced by fractional seconds, e.g. use **%H:%M:%S.###** for the time to the nearest millisecond. Sub-second times are only approximate and currently decimal places after the third will always be reported as zero.

- **%D**

The current UTC date and time in ISO 8601 format (**YYYY-MM-DD HH:MM:SS**).

- **%D{format}**

The current UTC date and time in the specified *format*, which takes the same format as the **template** argument to **strftime(3)**. Supports the same extension for sub-second resolution as **%d{...}**.

- **%E**

The hostname of the node running the application.

- **%m**

The message being logged.

- **%N**

A serial number for this message within this run of the program, as a decimal number. The first message a program logs has serial number 1, the second one has serial number 2, and so on.

- **%n**

A new-line.

- **%p**

The level at which the message is logged, e.g. **DBG**.

- **%P**

The program's process ID (pid), as a decimal number.

- **%r**

The number of milliseconds elapsed from the start of the application to the time the message was logged.

- **%t**

The subprogram name, that is, an identifying name for the process or thread that emitted the log message, such as **monitor** for the process used for **--monitor** or **main** for the primary process or thread in a program.

- **%T**

The subprogram name enclosed in parentheses, e.g. **(monitor)**, or the empty string for the primary process or thread in a program.

- **%%**

A literal **%**.

A few options may appear between the **%** and the format specifier character, in this order:

- **-**

Left justify the escape's expansion within its field width. Right justification is the default.

- **0**

Pad the field to the field width with **0** characters. Padding with spaces is the default.

- **width**

A number specifies the minimum field width. If the escape expands to fewer characters than *width* then it is padded to fill the field width. (A field wider than *width* is not truncated to fit.)

The default pattern for console and file output is **%D{%Y-%m-%dT%H:%M:%SZ}|%05N|%c|%p|%m**; for syslog output, **%05N|%c|%p|%m**.

Daemons written in Python (e.g. **ovs-monitor-ipsec**) do not allow control over the log pattern.

- **vlog/set FACILITY:facility**

Sets the RFC5424 facility of the log message. *facility* can be one of **kern**, **user**, **mail**, **daemon**, **auth**, **syslog**, **lpr**, **news**, **uucp**, **clock**, **ftp**, **ntp**, **audit**, **alert**, **clock2**, **local0**, **local1**, **local2**, **local3**, **local4**,

**local5, local6 or local7.**

- **vlog/close**

Causes the daemon to close its log file, if it is open. (Use **vlog/reopen** to reopen it later.)

- **vlog/reopen**

Causes the daemon to close its log file, if it is open, and then reopen it. (This is useful after rotating log files, to cause a new log file to be used.)

This has no effect if the target application was not invoked with the **--log-file** option.

## OPTIONS

- h, --help**

Prints a brief help message to the console.

- V, --version**

Prints version information to the console.

## SEE ALSO

**ovs-appctl** can control all Open vSwitch daemons, including **ovs-vswitchd(8)** and **ovsdb-server(1)**.

## AUTHOR

The Open vSwitch Development Community

## COPYRIGHT

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